Responsibility and verification of AI-generated code in online discussions

An exploratory study of five selected Hacker News discussions

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Code and data repository:
https://github.com/ledinhtrung05-coder/COMP3020-Group36

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1. Research purpose and design

1.1. Rationale

AI-assisted programming makes verification a substantive concern. A 24-participant study identified difficulties understanding and debugging generated suggestions (Vaithilingam et al. 2022). Barke et al. (2023) distinguished examination, execution, static analysis and documentation lookup. These concepts motivate examining verification language; they do not validate our dictionary or establish practice prevalence.

Here, **responsibility** means expressed expectations about who should understand, check, approve or remain answerable for code. GitHub's provider guidance recommends reviewing and validating suggestions (GitHub, n.d.). Reported confidence can nevertheless differ from observed verification success (O'Brien 2025). We therefore study expressed discussion and reply activity, not actual workplace behaviour or code quality.

Related discourse research includes Agarwal et al. (2026), a preprint proposing an explanatory theory whose causal propositions remain untested. Our contribution is a bounded connection between text, an exploratory association, lexical clusters and observed reply relationships, rather than a claim of discovering a new phenomenon.

1.2. Connected research questions

Overall question: How are responsibility and verification of AI-generated code discussed online, and how are these discussions structured through reply interactions?

Question	Analytical contribution
RQ1. What responsibilities and verification practices are discussed in relation to AI-generated code?	Describe words and candidate mentions; interpret examples in context.
RQ2. Is mentioning at least one specific code-verification practice associated with whether a comment receives a direct reply?	Test a comment-level association using the teammate's supplied labels.
RQ3. What content clusters emerge, and which aspects of responsibility or verification characterise them?	Assess whether lexical groups support meaningful themes.
RQ4. Which participants and content clusters receive replies, and how are reply interactions structured?	Analyse the author network and connect replies to receiving-comment clusters.

Table 1: Connected research questions and analytical contributions

RQ1 motivates examining practice mentions and replies in RQ2; RQ3 checks thematic structure before RQ4 links content with interactions. RQ2 replaces the earlier reading-versus-testing proposal with the teammate's completed question. RQ4 retains the teammate's structural analysis and adds the content bridge. These revisions are explicit; the design was not preregistered.

2. Data, collection and analytical boundaries

2.1. Source and collection provenance

The group confirms collecting the data using R on **28 Tháng Chín 2026**; the exact time and timezone are unavailable. The recovered collector uses jsonlite::fromJSON and the official Hacker News item endpoint, traversing kids from five fixed story IDs. It retains IDs, parents, authors, Unix times, HTML text, availability flags and story titles. The API supports text analysis and reconstruction of observed reply links (Hacker News, n.d.).

Sample rationale. The five supplied discussions address independent review, verification costs, automated checking and human validation of AI-generated code. These topics connect the verification concerns in Vaithilingam et al. (2022) and Barke et al. (2023) to our questions about responsibility and replies. This explains the corpus's relevance; it is not a reconstruction of unrecorded search queries, screening counts or exclusion decisions. The sample is bounded and non-probability, not an exhaustive platform survey.

The archived collector and cleaner document implementation; Git dates are not collection dates. The original collector has no request log or retry mechanism, so missing items and later replies remain unobserved. The separate collection guide records each thread's relevance and source limitations. Knit uses the frozen CSVs; the optional collector creates a separate new snapshot.

```
raw <- read_comments(here::here("data", "raw", "hn_comments_raw.csv"))
comments <- read_comments(
  here::here("data", "processed", "hn_comments_clean.csv"), clean = TRUE)
audit <- audit_comments(raw, comments)
```

2.2. Cleaning audit and sample composition

The recovered cleaner removes dead/deleted comments, strips HTML tags, decodes specified entities, collapses whitespace and converts times to UTC. The integrated audit checks IDs, unchanged shared fields and the retained-reply definition without replacing the supplied clean text.

metric	value
Raw comments	313
Clean comments	278
Removed dead	30
Removed deleted	5
Threads	5
Clean authors	186
Authors with multiple comments	55

Table 2: Data and reply-definition audit

Removed IDs are exactly the dead/deleted records; retained shared fields are unchanged. Unique IDs and matched raw parents establish internal consistency, not collection completeness. The full audit is exported.

Thread ID	Thread title	Raw	Clean	Authors
43857643	AI code review: Should the author be the reviewer?	85	84	52
47289406	Verification debt: the hidden cost of AI-generated code	102	96	68
47397367	Toward automated verification of unreviewed AI-generated code	99	83	58
49321400	Ask HN: What tools are you using for human code review of AI-assis…	17	11	10
49378314	Ask HN: How do you review and validate LLM generated code?	10	4	4

Table 3: Selected discussions; author counts are not additive across threads

Comment timestamps span 01 Tháng Năm 2025 to 25 Tháng Tám 2026; this is not continuous coverage. The three largest threads contribute 94.6% of comments. Selection is not a probability sample of Hacker News or programmers.

2.3. Variables and units

Variable or object	Definition and unit
Comment, thread and parent IDs	Character keys; joins use IDs, never row positions.
text / text_raw	Clean display text / preserved HTML source.
received_reply	At least one direct child retained in the clean corpus.
verification_specific	Supplied specific-practice label; separate from the dictionary.
Candidate indicators	Non-exclusive lexical matches, not validated attitudes.
Text cluster	Assignment of an eligible comment, not an author's position.
Author edge	Replying author to the retained parent-comment author.

Table 4: Analytical variables and units

Using raw child IDs changes 1 retained comment's reply flag. We consistently use clean children. Removed, later or uncollected replies are unobserved, rather than proof that no reply ever occurred. Root stories are absent, so story-level replies do not become author edges.

3. Research question

RQ1: responsibility and verification in the text

Representation and descriptive measures

We lowercase ASCII text, remove URLs, expand common contractions and keep alphabetic tokens of at least two characters. A frozen English stopword list retains negators. Unstemmed words remain readable but split related forms. Quoted material is retained because flattened boundaries cannot reliably identify quote-only text.

```
text_result <- analyse_text(comments)
head(text_result$frequencies, 20)
```

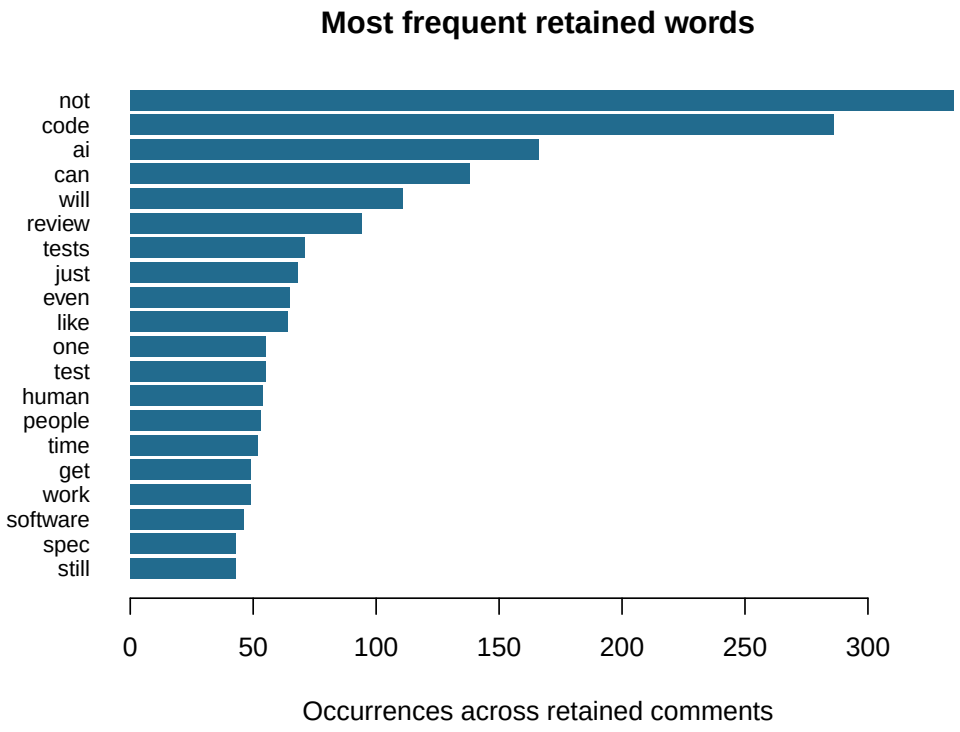


Figure 1: Frequent retained words. Counts include quotations and describe occurrences, not people or endorsement.

The leading terms are not, code, ai, can, will, review. Frequency identifies visible vocabulary, not responsibility, endorsement or effectiveness. Contextual validation remains necessary (Grimmer and Stewart 2013).

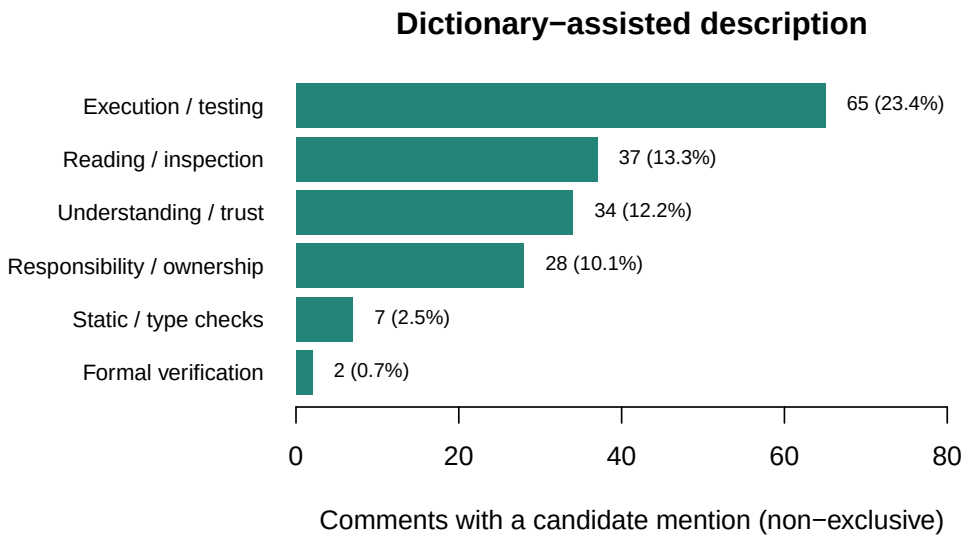


Figure 2: Comments matching each dictionary indicator. Categories overlap; matches may be quoted, negated or critical.

The indicators adapt verification concepts and add responsibility/formal-verification language. Exact patterns and matches are exported. These are screening counts: criticism can mention testing, while implicit practices may be missed. The dictionary does not create RQ2 labels.

Contextual reading: what the indicators can and cannot mean

Purposive readings cover all five discussions, contrasting claims and sarcasm. Twelve examples, including the original four, are exported with exact excerpts and full text. They illustrate meaning, not prevalence.

Comment IDs	Contextual interpretation
43862722 / 43867077	Human ownership supports accountability; imposing fixes on approvers may discourage review.
47292915	AI users should own review, qualified by a high-trust setting.
47416156	Engineers must choose tests using system knowledge.
49326866	Review tools may help, but missed defects and cost remain concerns.
49378613	Human involvement, coverage and separate test writing are proposed safeguards.
43862230	Sarcasm about regenerated tests warns against equating keywords with approval.

Table 5: Cross-thread examples and contrasting readings

These cases connect **who must check** with **what makes checking credible**. They motivate RQ2's practice-mention question and RQ4's distinction between interaction and agreement; they do not measure effectiveness.

RQ2: specific verification mentions and direct replies

Question, measurement and hypotheses

H0: specific-practice mentions and retained direct replies are independent. **H1:** they are associated, without a specified direction. The reference significance level is 0.05.

The group reports creating the labels using code in Excel. The source definition of TRUE is a mention of a specific way to check or test AI-generated code. The recovered R script assigns TRUE to 123 listed IDs and FALSE otherwise, exactly reproducing the supplied labels. The original Excel expression has not been inspected, so its handling of quotations and ambiguous cases is unverified. Labels are preserved; the dictionary is not their source.

```
rq2 <- analyse_hypothesis(comments,
  here::here("data", "annotations", "verification_labels.csv"))
chisq.test(rq2$table, correct = TRUE)
```

Specific practice	No reply	Reply	Total	Reply percent
FALSE	86	69	155	44.5
TRUE	71	52	123	42.3

Table 6: Specific verification mention and retained direct reply

All 278 labels join by comment_id, with shared fields checked. Comment 47290993 is FALSE despite discussing review; 43862196 is TRUE for a challenged quoted-review claim. These require rule clarification, not silent recoding. Matching the original table verifies arithmetic, not coding validity.

Test and interpretation

Two binary variables produce a 2-by-2 count table. Pearson's chi-square is the association test under an independent-comment reference model; Yates' correction is explicit (R Core Team, n.d.).

Specific practice	No reply	Reply
FALSE	87.54	67.46
TRUE	69.46	53.54

Table 7: Expected cell counts under independence

All expected counts exceed five (minimum 53.54), but this **does not establish independence**. There are 55 repeat-contribution authors and 5 nested discussions. Neither Fisher's test nor a Monte Carlo p-value would remove this dependence.

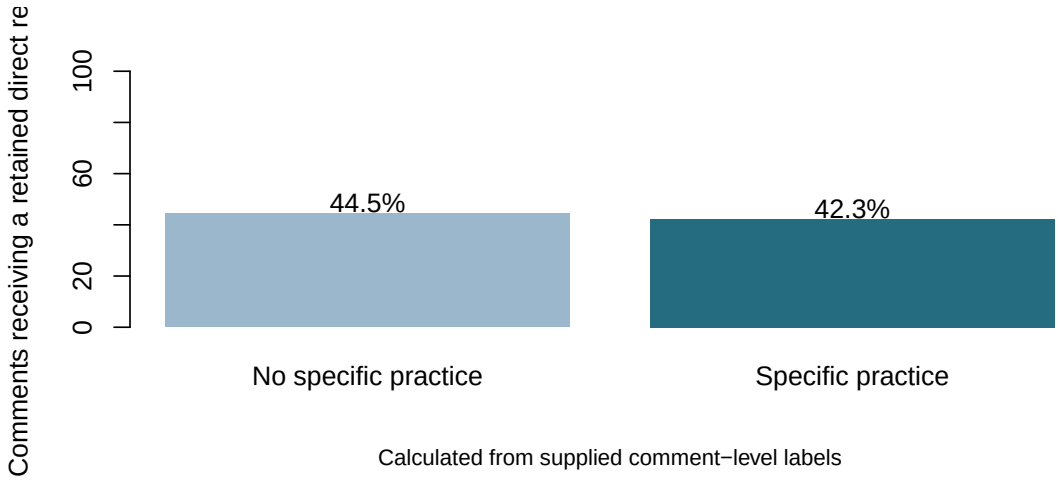


Figure 3: Observed retained-reply proportions from supplied labels. Denominators are comments; reply does not imply agreement.

The reference test gives χ^2 = 0.0637, df = 1, p = 0.8008. Rates are 42.3% for TRUE and 44.5% for FALSE: a -2.24 percentage-point difference.

The reference model does not reject H0. This is not evidence of equivalence or no relationship.

Effect size and design matter beyond a significance label (American Statistical Association 2016). Timing, visibility and text length are uncontrolled. This exploratory result is neither causal nor generalisable to all programmers.

Descriptive check by discussion

Thread	FALSE n	FALSE reply %	TRUE n	TRUE reply %
43857643	42	50.0%	42	47.6%
47289406	66	43.9%	30	40.0%
47397367	46	41.3%	37	48.6%
49321400	0	n/a	11	18.2%
49378314	1	0.0%	3	0.0%

Table 8: Reply proportions within each thread and supplied label; n/a means no comments in that category

One category is absent in 1 thread(s). Its rate is unavailable, not zero. Uneven denominators help explain the pooled composition; this breakdown does not adjust for dependence.

Sensitivity to authors and thread composition

Repeated observations can change weighting and violate independence (Altman and Bland 1997). As a post-hoc sensitivity, select each author's earliest retained comment by timestamp, breaking ties by numeric ID. Selection uses neither label nor reply outcome. Preserve its replies from the **full corpus**; recomputing replies within the reduced subset would wrongly change the outcome.

View	n	FALSE %	TRUE %	Diff. pp	Ref. p
All comments	278	44.516	42.276	-2.240	0.801
First per author	186	47.475	48.276	0.801	1.000

Table 9: Design sensitivity; both p-values retain an independence assumption

The reduced view removes repeat-author weighting but describes first retained comments, not all contributions. Its contrast is 0.80 percentage points. Shared-thread and interacting-author dependence remain. Omitting each thread in turn gives differences from -6.34 to 0.13 points; all five are exported. These checks show that the direction is not stable across these design choices. They are not a dependence-adjusted test or evidence of equivalence.

Sensitivity to the measurement rule

A separate AI-assisted full-text pass applies the frozen **AI-SENS-v1** rule: explicit checking activities count, including quotations, criticism and agent checks; generic verification alone does not. Decisions include exact excerpts and reasons. The reading view omitted labels/outcomes, but prior context exposed several labelled examples, so this is **not independent blinded or human validation** (Grimmer and Stewart 2013). The completed audit was frozen before comparison and never replaces the teammate's labels.

Of 278 readings, 5 remain ambiguous. Among 273 determinate cases, 14 differ from supplied labels. This is disagreement between operationalisations, not a coding-error rate. For example, 47290993 explicitly mentions reviewing code, whereas 47290312 says only to prompt AI to verify software.

Rule	n	FALSE %	TRUE %	Diff. pp
Supplied	278	44.52	42.28	-2.24
AI rule: determinate	273	43.45	44.53	1.08

Table 10: Descriptive measurement sensitivity; the AI rule is not a gold standard

Assigning all ambiguous cases FALSE or all TRUE gives contrasts of 1.86 and 0.16 points. These are declared scenarios, not mathematical bounds. Measurement choice also changes the contrast's direction; AI interpretation errors and dependence remain. No further significance tests are selected from these views.

RQ3: exploratory content clusters

Representation, distance and selection rule

For clustering only, broad terms ai, code, llm, llms, model, models, generated, generative are removed. Comments need five retained tokens; vocabulary must occur in at least two documents and at most 80% of eligible documents. Empty vectors are excluded. These pragmatic cutoffs limit sparse features; excluded comments remain in the reply network.

Nonzero TF is $1 + \log(c_{dt})$, IDF is $\log(N/df_t)$, followed by L2 normalisation. Cosine dissimilarity $1 - x_i^T x_j$ compares vocabulary profiles with reduced raw-length effects (Manning et al. 2008). Authors, threads and reply outcomes are not input features.

```
clusters <- analyse_clusters(comments, text_result)
clusters$sensitivity <- cluster_linkage_sensitivity(clusters)
```

Average linkage groups comments by mean pairwise dissimilarity. Among k = 2–8, we maximise mean silhouette subject to at least five comments per cluster. If none passes, the highest silhouette is a **diagnostic fallback**, not a validated optimum. Silhouette contrasts within-cluster and nearest-other-cluster dissimilarity (R Core Team and cluster package authors, n.d.).

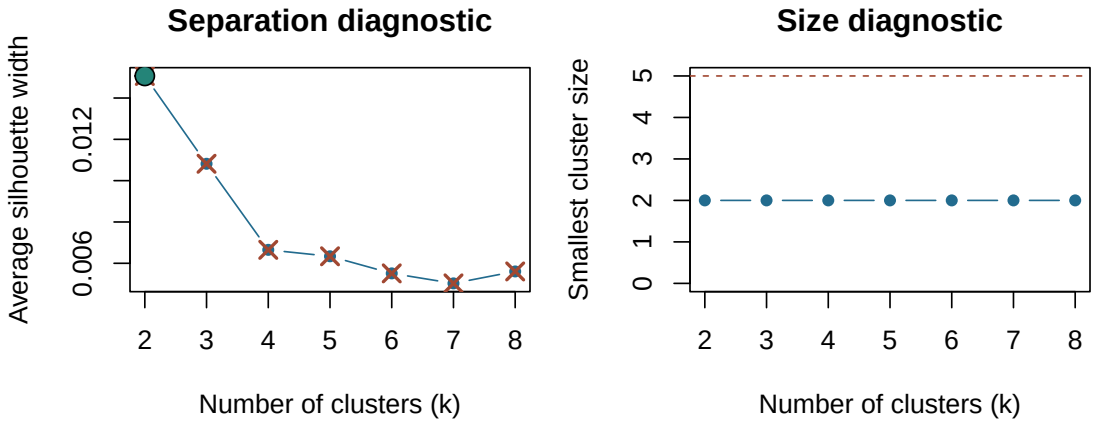


Figure 4: Separation and minimum-size diagnostics. Crosses fail the five-comment guard; the filled marker shows the declared selection.

There are 264 eligible comments and 1250 terms; 14 comments are unassigned. Selected k = 2, mean silhouette = 0.0151. No tested k had all clusters of at least five documents; highest silhouette fallback.

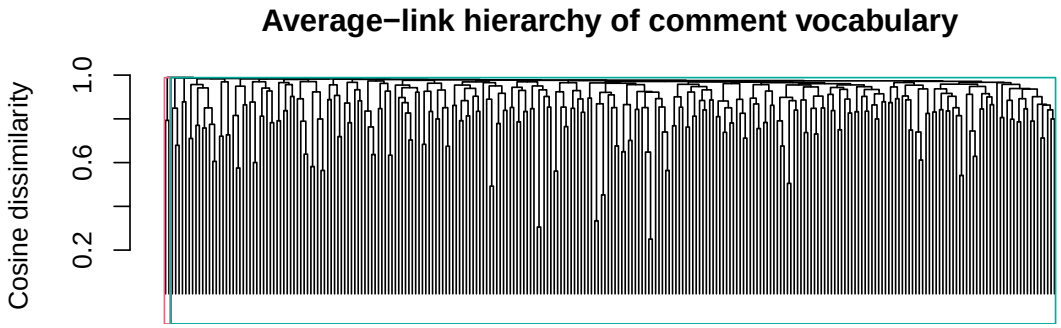


Figure 5: Average-link hierarchy with selected cut. Rectangles indicate lexical partitions, not social groups.

Cluster profiles and contextual checks

Group	n	Mean silhouette	Median tokens	Top terms
1	262	0.014	27	not, will, can, review, tests, like
2	2	0.199	12	hand, driven, design, custom, phenomenon, rare

Table 11: Exploratory profiles; top terms are descriptors, not validated theme names

The medoid minimises mean within-cluster dissimilarity; the boundary example has the lowest silhouette. Their complete texts and IDs are exported for inspection.

Cluster 1 is a heterogeneous remainder. Its medoid (47290376) questions specifications and unit tests while discussing simulation and human review. Boundary comment 47295857 advocates test-driven development. Shared review/testing words therefore cover different arguments.

Cluster 2 is a two-comment lexical pairing. Comment 43877106 links specification-driven and test-driven design as ‘hand in hand’; 47289697 contrasts disposable generated code with ‘hand crafted’ code. The common word *hand* has different meanings. This is not a coherent verification theme.

The primary representation does not support well-separated, balanced thematic clusters. We report a diagnostic partition without inventing stance labels.

Median length, thread composition and complete examples remain exported. They diagnose interpretation rather than establish semantic validity. Relevance was not manually coded, so contextual or promotional material can remain.

Sensitivity and diagnosis

Complete linkage retains cosine distance; Ward.D2 uses Euclidean distance on L2-normalised vectors. Both retain the k range and size guard.

Linkage	k	Silhouette	Sizes	Size rule
Average	2	0.0151	262/2	Fail
Complete	2	-0.0020	259/5	Pass
Ward.D2	2	0.0025	241/23	Pass

Table 12: Selected diagnostic partition within each linkage/distance specification

Both alternatives have near-zero selected silhouette and do not support clear themes. Different distance scales prevent treating this as an algorithm leaderboard; internal fit also does not establish semantic validity (Manning et al. 2008).

The matrix is 97.9% zero. Median cosine distance is 0.990; 31.9% of pairs share no retained term. Sparse overlap helps explain weak lexical separation.

Setting	n	Terms	Sizes	Silhouette
Primary	264	1250	262/2	0.0151
Min. 10 tokens	227	1223	226/1	0.0157
Min. 20 tokens	167	1127	165/2	0.0184
Min. term df 3	264	767	262/2	0.0196

Table 13: Declared post-hoc representation checks using the same average-link selection rule

All three alternatives address short documents or rare terms; none replaces the primary result. Their low separation strengthens the diagnosis of these lexical representations, without proving that the discussions contain no themes.

RQ4: direct-reply structure and the content bridge

Network construction

Nodes are authors participating in retained between-author replies. A directed edge A to B means A replied to B's comment. Repeated ordered pairs collapse into one edge with separate event frequency. Self-replies are excluded (0 here).

```
network <- analyse_network(comments, seed = 3020)
igraph::degree(network$graph, mode = "in", loops = FALSE)
igraph::betweenness(network$graph, directed = TRUE,
                    weights = NA, normalized = TRUE)
```

Metric	Value
Network authors	144
Unique directed author pairs	193
Density	0.0094
Weak components	9
Largest weak component	128
Strong components	102

Table 14: Directed author network properties

The graph combines 198 matched events. It excludes 42 other clean-corpus authors, 78 story replies and 2 replies to unavailable parents. Adding excluded authors as isolates would change density and normalised centrality.

Whole network and justified subgraph

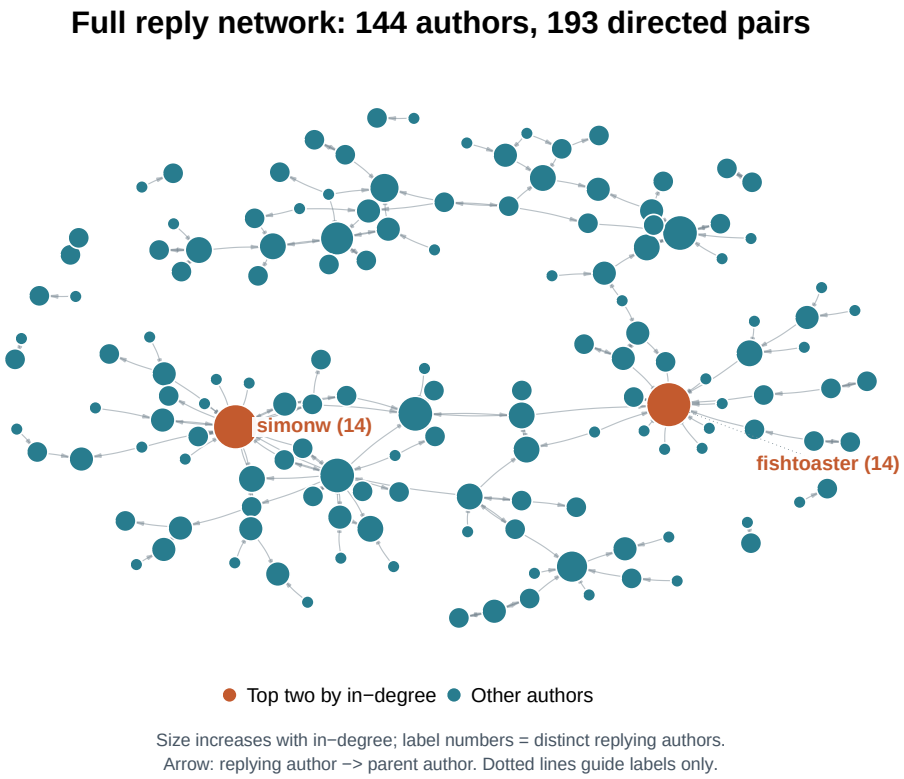


Figure 6: Full author reply network. Node size increases with square-root in-degree; orange highlights the two leading reply targets. Arrows point to the parent author; dotted lines guide labels.



Figure 7: Largest weak component. Node sizes use full-network in-degree; labels identify six leading reply targets. Seeded positions do not encode semantic similarity.

The largest weak component contains 128 authors (88.9% of the graph). The largest strong component has only 10, so weak connectivity is not reciprocal exchange. Pooling threads can connect repeat participants without making them one conversation or establishing consensus. The degree distribution is exported.

Centrality and interpretation

In-degree counts distinct replying authors; incoming strength counts reply events. Betweenness measures directed shortest-path positions, normalised across the full graph. It is **unweighted**: reply frequency is not a distance (igraph contributors, n.d.).

Author	In-degree	Reply events received	Normalised betweenness
simonw	14	14	0.0191
fishtoaster	14	15	0.0030
dakshgupta	7	7	0.0141
logicchains	7	7	0.0089
apical_dendrite	7	7	0.0027
Swizec	6	6	0.0033
jryio	5	5	0.0005
duskdozer	4	4	0.0000

Table 15: Participants ranked by in-degree, then betweenness

simonw and fishtoaster lead in-degree (14 and 14). Their betweenness values are 0.0191 and 0.0030: receiving attention and linking directed paths are different roles. Neither proves expertise, agreement or causal influence. These calculations reproduce the teammate PDF's principal numerical findings; reconstructed code and originals remain separately identified.

Context clarifies prominence. simonw's responsibility claim (43861845) receives 6 retained direct replies. fishtoaster's proposed verification safeguards (47290195) receive 13, while acknowledging they may be insufficient. Reply 47290993 challenges specification-only review; 47292581 gives qualified support with a task-scope condition. These are illustrative readings, not sentiment counts. Their actual parent-child links connect RQ1's contested verification language with network structure and show why RQ2's reply flag measures interaction, not approval.

Which content receives replies?

Clusters join to **receiving parent comments** by comment_id. An author can write across clusters. Unassigned child comments still count as replies to an assigned parent.

```
bridge <- analyse_cluster_replies(comments, clusters$assignments)
```

Receiver cluster	Comments	Comments with reply	Reply percent	Incoming reply events
1	262	118	45	195
2	2	0	0	0

Table 16: Descriptive reply outcomes by receiving-comment cluster

This connects content and interaction without assigning one theme to each author. Weak clustering limits its substantive meaning: a rate based on only two lexically paired comments cannot establish whether a theme attracts replies. No extra significance test is performed. The exported sender-by-receiver matrix requires both labels and reports excluded-event coverage.

4. Overall findings, limitations and conclusion

Across the five selected discussions, examples attach accountability to people while proposed checking work shifts among code review, specifications, test design and automated tools. Replies include challenges and qualified support, not a demonstrated consensus. Specific-practice mentions receive retained replies at 42.3%, versus 44.5% otherwise (difference -2.24 percentage points; reference p = 0.8008). The first-comment-per-author contrast changes direction (0.80 points). Neither design supports a stable directional or causal conclusion about verification language and attention.

Clustering selects k = 2 with sizes 262/2 and mean silhouette 0.0151. The partition does not establish distinct, balanced themes, restricting interpretation of cluster reply rates. Meanwhile, the observed network is sparse (density 0.0094), yet 128 of 144 authors share its largest weak component. Attention concentrates on particular reply targets, without implying influence or consensus.

The conclusion concerns **expressed responsibility and retained interactions**, not the best practice or code quality. Five selected, uneven threads restrict generalisation; exact search and request logs are unavailable. The uninspected Excel rule, ambiguous labels, AI-assisted reading uncertainty, dependent comments and unequal time to receive replies limit measurement and inference. The separate explicit-rule contrast is 1.08 points, reinforcing sensitivity to measurement choices. Sparse vocabulary limits clustering; omitted parents and edge-only author selection constrain the network. Reviewed labels and broader documented sampling should precede stronger claims.

5. Reproducibility and group contributions

Relative paths, frozen UTF-8 files, character IDs, explicit logical parsing and UTC dates support consistent runs. Validation reloads disk inputs and rejects missing/mismatched IDs. No API request occurs during Knit. Installation and recollection are separate steps.

The teammate's RQ2 question, labels and test are preserved; RQ4's principal network results are reproduced, with added content linkage. Original submissions and recovered repository scripts are archived. New RQ1/RQ3 modules and integration decisions remain identifiable. The originals use historical paths and are evidence files, not scripts to source during the integrated run.

The repository includes complete functions, data, run instructions and exported tables/figures. Appendix A gives core operations; outputs/session_info.txt records the environment. Actual contributions belong in docs/group_contributions.csv.

AI assistance and group evaluation: AI supported source research, code development, draft review

Appendix A: core analytical operations

These excerpts show the central calculations. Full functions add the documented filters, joins and validation; versions are recorded in outputs/session_info.txt.

```
tf <- counts
tf[counts > 0] <- 1 + log(counts[counts > 0])
idf <- log(nrow(counts) / colSums(counts > 0))
tfidf <- sweep(tf, 2, idf, "*")
tfidf <- tfidf / sqrt(rowSums(tfidf^2))
d <- 1 - tcrossprod(tfidf)
d[d < 0] <- 0
diag(d) <- 0
tree <- hclust(as.dist(d), method = "average")
groups <- cutree(tree, k = selected_k)
cluster::silhouette(groups, as.dist(d))
```

```
# author_edges contains matched, non-self ordered pairs and event frequency
g <- igraph::graph_from_data_frame(author_edges, directed = TRUE)
igraph::degree(g, mode = "in", loops = FALSE)
igraph::betweenness(g, directed = TRUE, weights = NA, normalized = TRUE)
```

```
first <- order(comments$time, as.numeric(comments$comment_id))
first <- first[!duplicated(comments$author[first])]
rq2$joined[first, ]  # retain reply outcomes from the full corpus
rq2$sensitivity$summary
label_sensitivity$summary  # separate frozen AI-assisted rule
```

References

Agarwal, Shyam, Courtney Miller, Christian Kästner, and Bogdan Vasilescu. 2026. *3100 Opinions on Code Review in an AI World: Building Causal Theory from Practitioner Discourse*. https://doi.org/10.48550/arXiv.2607.07980.

Altman, Douglas G., and J. Martin Bland. 1997. "Statistics Notes: Units of Analysis." *BMJ* 314 (7098): 1874. https://doi.org/10.1136/bmj.314.7098.1874.

American Statistical Association. 2016. *American Statistical Association Releases Statement on Statistical Significance and p-Values*. https://www.amstat.org/asa/files/pdfs/p-valuestatement.pdf.

Barke, Shraddha, Michael B. James, and Nadia Polikarpova. 2023. "Grounded Copilot: How Programmers Interact with Code-Generating Models." *Proceedings of the ACM on Programming Languages* 7 (OOPSLA1). https://doi.org/10.1145/3586030.

GitHub. n.d. *Application Card: GitHub Copilot Inline Suggestions*. Accessed October 9, 2026. https://docs.github.com/en/copilot/responsible-use/inline-suggestions.

Grimmer, Justin, and Brandon M. Stewart. 2013. "Text as Data: The Promise and Pitfalls of Automatic Content Analysis Methods for Political Texts." *Political Analysis* 21 (3): 267–97. https://doi.org/10.1093/pan/mps028.

Hacker News. n.d. *Hacker News API*. Accessed October 9, 2026. https://github.com/HackerNews/API.

igraph contributors. n.d. *Vertex and Edge Betweenness Centrality*. Accessed October 9, 2026. https://r.igraph.org/reference/betweenness.html.

Manning, Christopher D., Prabhakar Raghavan, and Hinrich Schütze. 2008. *Introduction to Information Retrieval*. Cambridge University Press. https://nlp.stanford.edu/IR-book/.

O'Brien, Gabrielle. 2025. "How Scientists Use Large Language Models to Program." *Proceedings of the CHI Conference on Human Factors in Computing Systems*. https://doi.org/10.1145/3706598.3713668.

R Core Team. n.d. *Pearson's Chi-Squared Test for Count Data*. Accessed October 9, 2026. https://stat.ethz.ch/R-manual/R-devel/library/stats/html/chisq.test.html.

R Core Team and cluster package authors. n.d. *Compute or Extract Silhouette Information from Clustering*. Accessed October 9, 2026. https://stat.ethz.ch/R-manual/R-devel/library/cluster/html/silhouette.html.

Vaithilingam, Priyan, Tianyi Zhang, and Elena Glassman. 2022. "Expectation Vs. Experience: Evaluating the Usability of Code Generation Tools Powered by Large Language Models." *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. https://doi.org/10.1145/3491101.3519665.